

“The standards used to measure safety in autonomous cars will be far more severe than for humans,” says Rojas. “But once all pieces fall into place, autonomous cars will be safer than cars driven by persons. It is just a matter of time”.

Aircraft that learns by watching other aircrafts

The war in Afghanistan brought to light the advances in unmanned aerial vehicles (UAV) like the Predator and the Reaper, but this kind of overshadowed what scientists are doing with autonomous UAVs. The technologies employed in autonomous cars is similar to what is used for aircrafts, but one research group in Stanford University is really pushing the boundaries of artificial intelligence.

The computer scientists have developed an artificial intelligence system that enables robotic helicopters to teach themselves to fly difficult stunts by watching other helicopters perform the same manoeuvres. The result is an autonomous helicopter that can perform a complete airshow of complex tricks on its own.

The stunts are “by far the most difficult aerobatic manoeuvres flown by any computer controlled helicopter,” said Andrew Ng, the professor directing the research of graduate students Pieter Abbeel, Adam Coates, Timothy Hunter and Morgan Quigley.

The dazzling airshow is an important demonstration of “apprenticeship learning”, in which robots learn by observing an expert, rather than by having software engineers peck away at their keyboards in an attempt to write instructions from scratch.

Stanford’s artificial intelligence system learned how to fly by “watching” the four-foot-long helicopters flown by expert radio control pilot Garrett Oku. “Garrett can pick up any helicopter, even ones he’s never seen, and go fly amazing aerobatics. So the question for us is always, why can’t computers do things like this?” Coates said.

Computers can, it turns out. One morning in an empty field at the edge of campus, Abbeel and Coates sent up one of their helicopters to demonstrate autonomous flight. The aircraft is an off-the-shelf radio control helicopter, with instrumentation added by the researchers.

For five minutes, the chopper, on its own, ran through a dizzying series of stunts beyond the capabilities of a full-scale piloted helicopter and other autonomous remote control helicopters. The artificial-intelligence helicopter performed several difficult manoeuvres.